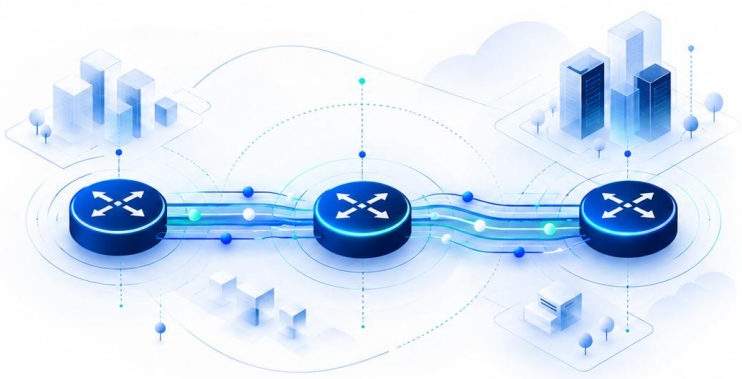
AIORI HACKATHON 3

A3-PS008-TC001

Network Management Agent for Reviewed Configuration Changes

Intent-Based Network Configuration with Pre-Deployment Validation, Human Approval and Recovery



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A safer way to change network configurations

How can a proposed network change be validated and reviewed before it modifies a router?

Abstract

This problem statement proposes a network management agent that converts a supported network change request into a candidate configuration, evaluates its expected effect, and presents the evidence for human review. After approval, the agent applies the permitted change, verifies the resulting network state and attempts recovery (as per scope) if execution fails.

Problem

- Manual router configuration can introduce incorrect prefixes, unreachable next hops, routing conflicts and connectivity failures.
- A command being accepted by a router does not guarantee that the intended network outcome has been achieved.

Illustrative request - “Make the network 10.20.0.0/24 reachable from R1.”

01 Validate the request

Accept only supported IPv4 static-route add/remove requests.

02 Assess

Check the candidate against observed routes and the modeled topology.

03 Review

Display the target router, exact command, expected impact and evidence.

04 Verify the result

Read back the resulting route and run a configured connectivity probe for additions.

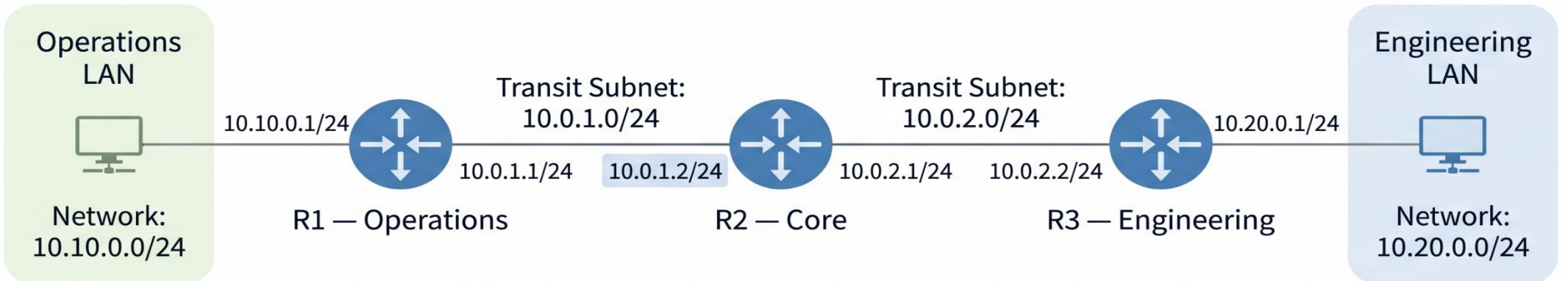
APPROACH

Intent → Validate → Preview → Human Approval → Deploy → Verification/Recovery

Current scope: IPv4 static routes | Three router isolated lab | No arbitrary CLI or production router access

Start with the network the change affects

Three router lab makes the route decision visible to reviewers.



Candidate static route on R1: 10.20.0.0/24 via 10.0.1.2

Three router testbed

R1 - Operations: LAN 10.10.0.0/24
R2 - Core: Transit links to R1 and R3
R3 - Engineering: LAN 10.20.0.0/24

Example destination: 10.20.0.0/24 Example next hop from R1: 10.0.1.2 on R2

Subnet/Prefix

The destination network, such as 10.20.0.0/24.

Next hop

The neighboring router to which packets are forwarded.

Routing table

The routes a router uses to select a path.

Scenario: R1 has no suitable route to the Engineering LAN. We have to evaluate a candidate route through R2 before presenting it for approval.

Relevant Standards and Tools

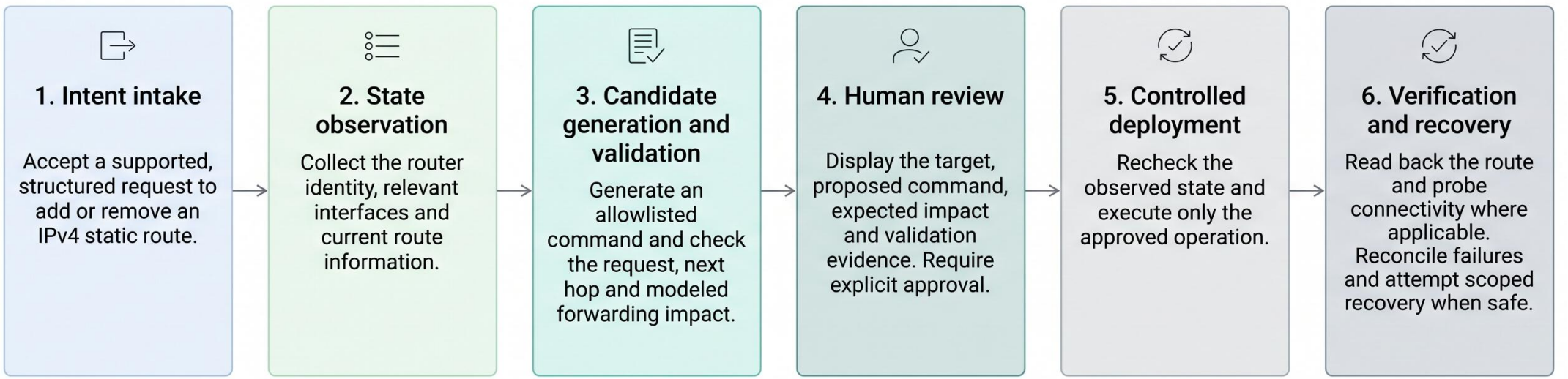
Reference	What it contributes	Relevance to problem statement
RFC 1812 [1]	IPv4 router behaviour and forwarding concepts.	Provides the basis for reasoning about forwarding and next hops.
RFC 4632 [2]	Classless Inter-Domain Routing (CIDR) prefixes and route aggregation.	Supports correct interpretation of IPv4 prefixes.
RFC 6241 - NETCONF [3]	Network configuration protocol, including candidate configuration and commit workflows	Provides a useful configuration management reference. The current adapter uses SSH CLI
GNS3 documentation [4]	Emulated network topology and controller architecture	Provides the isolated lab environment and topology-management interface.
Netmiko [5]	Python library for device interaction over SSH	Provides the CLI transport used by the IOS compatible adapter.

Key distinction: GNS3 runs the lab & the problem statement solution supplies the change review agent workflow.

Research gap addressed: Device access and routing standards do not, by themselves, implement this prototype's complete application level workflow for impact review, explicit human approval, state rechecking and scoped recovery.

A review-first network configuration agent

The same review lifecycle serves the deterministic software twin and the separate GNS3 device adapter.



Guardrails

- ✓ Only supported route operations are accepted
- ✓ No device write before approval
- ✓ Recheck state before deployment
- ✓ Route removal is restricted to changes owned by the agent
- ✓ Uncertain device state stops further automation and requires recovery review

Current implementation: Deterministic request handling; LLM-based CLI generation is deferred.

From manual configuration to evidence-based change control

Comparison describes the intended workflow improvement

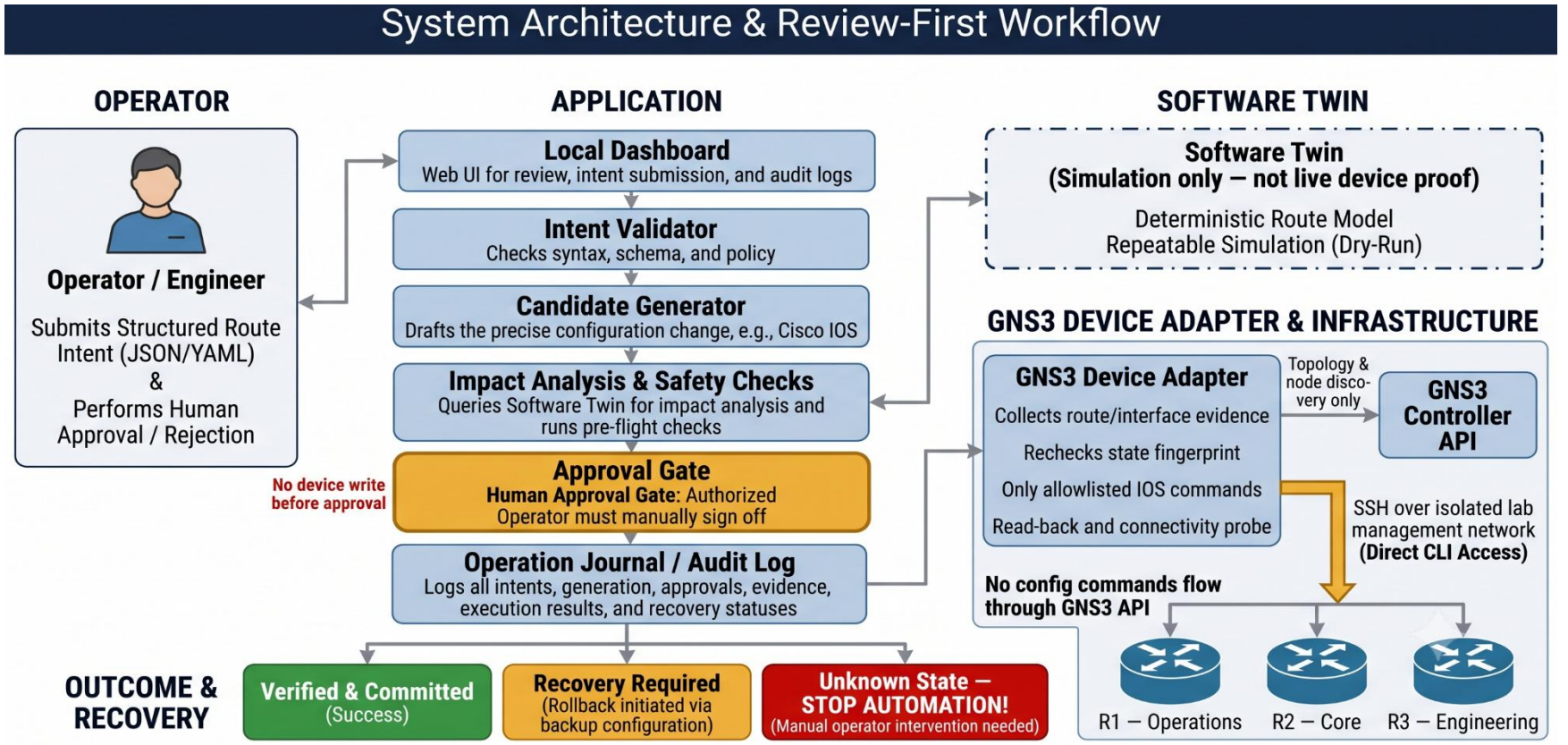
Risk / step	Conventional manual workflow	PS-008 approach	Evidence status
Incorrect request	Errors may reach the command entry stage	Typed IPv4 add/remove request; strict validation.	Implemented
Unexpected route impact	Operator must reason about forwarding manually	Preview before/after forwarding across observed sources.	Software twin / adapter scope
Unauthorized change	Approval may be informal or separate	Review-bound, single-use explicit approval.	Implemented
Stale device state	Command can be issued without app-level checks. Network may change after review	Allowlisted IOS command after re-observation and approval.	Mock tested
False success	Command acceptance may be mistaken for success	Route read-back and configured probe for additions	Live proof pending
Failed deployment	Retry or rollback may be ambiguous.	Reconcile timeout; scoped recovery; unknown state blocks.	Mock tested

Agent boundary: propose → explain → verify. The operator remains the authority for deployment.

Optimization goal: Make each change reviewable, each deployment conditional on approval, and each outcome evidence-based.

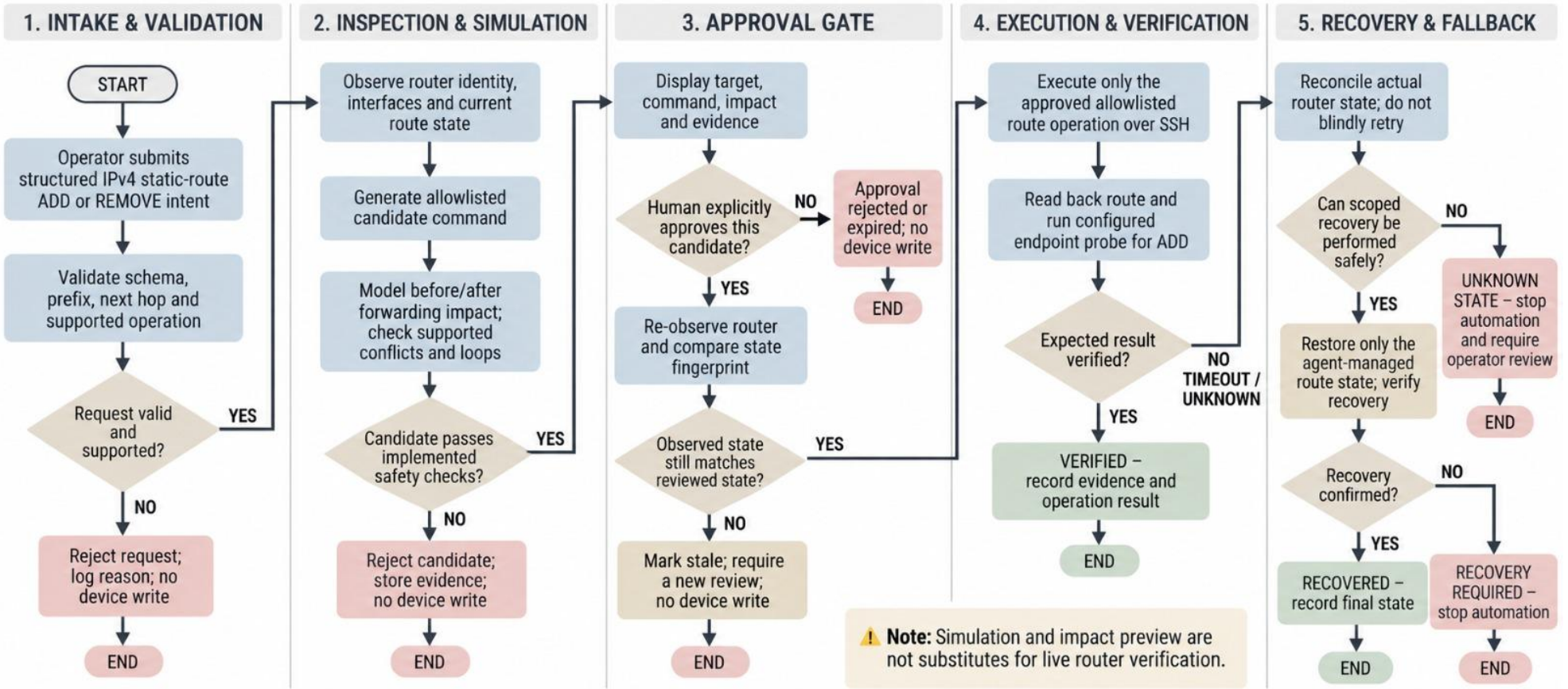
Shared safety workflow, separate execution backends

The review layer is shared by the software twin and the GNS3 device adapter. The twin supports repeatable simulation; the adapter interacts with the configured lab routers after approval.



Key distinction: GNS3 provides the lab environment and topology API. Proposed solution performs the review and safety checks; the device adapter uses SSH to interact with the configured routers. The software twin is a separate simulation backend.

From Operator Intent (change) to Verified Result



Reject / stale / unsupported output → no write

Timeout or mismatch → reconcile; restore only managed route; otherwise recovery_required / unknown

Fail closed rule: Invalid, rejected, unsupported or stale changes must not reach the device-write stage. If a deployment times out or its outcome is uncertain, reconcile the router state before taking further action.

Current prototype and path to complete validation

Milestones are expressed as evidence gates rather than invented calendar dates.

Milestone	What is ready	Exit evidence / next action
1 Software twin	Route preview, approval, persistence and simulator workflow.	Retain test output and demonstrate representative passing cases.
2. Safety and approval workflow	Request validation, review and approval flow implemented; selected behaviours mock tested	Demonstrate invalid requests, rejection, stale state and prevention of unauthorized writes.
3. GNS3 lab setup	GNS3 2.2.61 installed; router image and project topology pending at last check	Build the isolated three router topology using a permitted, compatible image.
4. Live device validation	Not yet demonstrated on a real device.	Test route addition/removal, route read-back, endpoint connectivity and failure handling on the configured lab.
5. Complete failure-recovery evidence	Recovery logic has offline/mock evidence; live behaviour pending	Test a failed or uncertain operation and verify whether recovery succeeds or the system stops safely.
6. Future enhancement: language interface	Deferred; no OpenAI call in current build	Consider after device workflow, credential handling and safety tests are stable.

Next validation gate: Completing the isolated R1–R2–R3 lab, verify router identity and SSH host keys, then capture evidence for an approved route addition, route removal, connectivity verification and a controlled failure scenario.

References

Primary standards and implementation references.

[1] **RFC 1812 — Requirements for IP Version 4 Routers**
https://www.rfc-editor.org/info/rfc1812/

[2] **RFC 4632 — CIDR Address Strategy**
https://www.rfc-editor.org/info/rfc4632/

[3] **RFC 6241 — Network Configuration Protocol (NETCONF)**
https://www.rfc-editor.org/info/rfc6241/

[4] **GNS3 Documentation — Architecture**
https://docs.gns3.com/docs/using-gns3/design/architecture

[5] **Netmiko — Python library for network-device interaction over SSH.**
https://github.com/ktbyers/netmiko

[6] **AIORI-3 Hackathon, "PS-008: Network Management Agent for Reviewed Configuration Changes," provided problem statement (detailed pdf)**

NETCONF is cited as a related device configuration standard; PS-008 currently uses SSH CLI.



Thank You

Network changes should be intentional, validated, approved and recoverable.

See the impact. Review the change. Verify the outcome.

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